

Modern Institute of Technology & Research Centre, Alwar

Assignment-3

Branch: AI&DS

Semester: VI

Subject: SMF (6AD4-03)

Faculty: Pradeep Kumar

Max. Marks: 30

Date of Issue: 21-02-2026

Last Date of Submission: 28-02-2026

Q1. A time series follows an AR(1) model given by

$$Y_t = 0.7Y_{t-1} + \varepsilon_t$$

where ε_t is white noise with mean zero. If $Y_{10} = 50$, compute the one-step, two-step, and three-step ahead forecasts. Also compute the theoretical autocorrelation at lag 1 and lag 2.

Q2. A time series becomes stationary after first differencing and follows an ARMA(1,1) model:

$$\Delta Y_t = 0.6\Delta Y_{t-1} + \varepsilon_t + 0.4\varepsilon_{t-1}$$

Given $\Delta Y_{20} = 5$ and $\varepsilon_{20} = 2$, compute the one-step ahead forecast for Y_{21} .

Q3. Consider quarterly seasonal data modeled by a SARIMA model:

$$SARIMA(1,1,1)(1,1,0)_4$$

(a) Write the full model equation using backshift operator notation.

(b) If the AIC values of three competing models are 520, 505, and 498, determine the best model using model selection criteria.

(c) Briefly explain how Impulse Response Function (IRF) can be used to compare two competing VAR models.

Q4. Explain Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) and describe how they are used to identify AR, MA, and ARMA models.

Q5. Define Vector Auto-Regressive (VAR) models and explain the concepts of Co-integration, Error Correction Models (ECM), Vector ARIMA models, and Impulse Response Function (IRF). Discuss how model selection criteria such as AIC and BIC help in finding the best model and how IRF is used to study differences between competing VAR models.